C++ Midway Project

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Explanation of program:

My command-line program is a functional book tracker for keeping track of books and their details. The user can add a book, define its title, author, genre, page count, and reading status, view all books, search for books tracked, and remove books. The list of books and data can be saved to a text file (`books.txt`) and loaded back again later so that data isn't lost when the program is exited.

My program uses two classes. `Book` stores a book's information, and `Library` stores all the books in a sorted array. Because the array is sorted by title, searching uses a binary search. The menu repeats until the user chooses to exit, and any invalid input shows an error instead of crashing the program.

Concepts Used:

- Data types and sizes:
  - `int`, `char`, `string`, `bool`, `float`, `double`, `long long`
  - sizes printed with `sizeof` in menu option 7
- File types and file I/O
  - `books.txt` (text file)
  - `ofstream`, `ifstream`
- Pointers:
  - `Book* books`
  - `Book* search()`
  - `->` operator
- Arrays:
  - Dynamic array of `Book` objects
  - `new` to create it
  - `delete[]` to free it
- Binary search:
  - `findIndex()`
- Strings:
  - title, author, genre
- Classes:
  - `Book`
  - `Library`

Screenshots:

```
=== Book Tracker ===
1. Add book
2. View all books
3. Search by title (binary search)
4. Remove book
5. Save to file
6. Load from file
```

Main menu

```
Choose an option: 1
Title: The Stranger
Author: Albert Camus
Genre: Classic
Pages: 144
Status (U=Unread, R=Reading, D=Done): R
Book added.
```

Adding a book

```
Choose an option: 2

#1
Title:  The Stranger
Author: Albert Camus
Genre:  Classic
Pages:  144
Status: Reading
```

Viewing books

```
Choose an option: 3
Title to search for: The Stranger

Found:
Title:  The Stranger
Author: Albert Camus
Genre:  Classic
Pages:  144
Status: Reading
```

Searching for a book that exists

```
Choose an option: 3
Title to search for: Pride and Prejudice
Book not found.
```

Searching for a book that doesn't exist

```
Choose an option: 4
Title to remove: The Stranger
Book removed.
```

Removing a book

```
Choose an option: 5
Saved 1 book(s) to books.txt
```

```
1    The Stranger|Albert Camus|Classic|144|R
2
```

Saving to file

```
Choose an option: 8
Goodbye!
```

Exiting

```
=== Book Tracker ===
1. Add book
2. View all books
3. Search by title (binary search)
4. Remove book
5. Save to file
6. Load from file
7. Show data type sizes
8. Exit
Choose an option: 6
Loaded 1 book(s) from books.txt

=== Book Tracker ===
1. Add book
2. View all books
3. Search by title (binary search)
4. Remove book
5. Save to file
6. Load from file
7. Show data type sizes
8. Exit
Choose an option: 2

#1
Title:  The Stranger
Author: Albert Camus
Genre:  Classic
Pages:  144
Status: Reading
```

Loading book from file after restarting program

```
Choose an option: 7

=== Data Type Sizes (bytes) ===
char:      1
bool:      1
int:       4
float:     4
double:    8
long long: 8
string:    32
Book:      104
Book*:     8 (a pointer)
```

Data type sizes

```
Choose an option: dfgf
Invalid input. Enter a number from 1 to 8.
Choose an option: 1
Title:
Invalid input. Cannot be empty or contain '|'.
```

Invalid inputs